

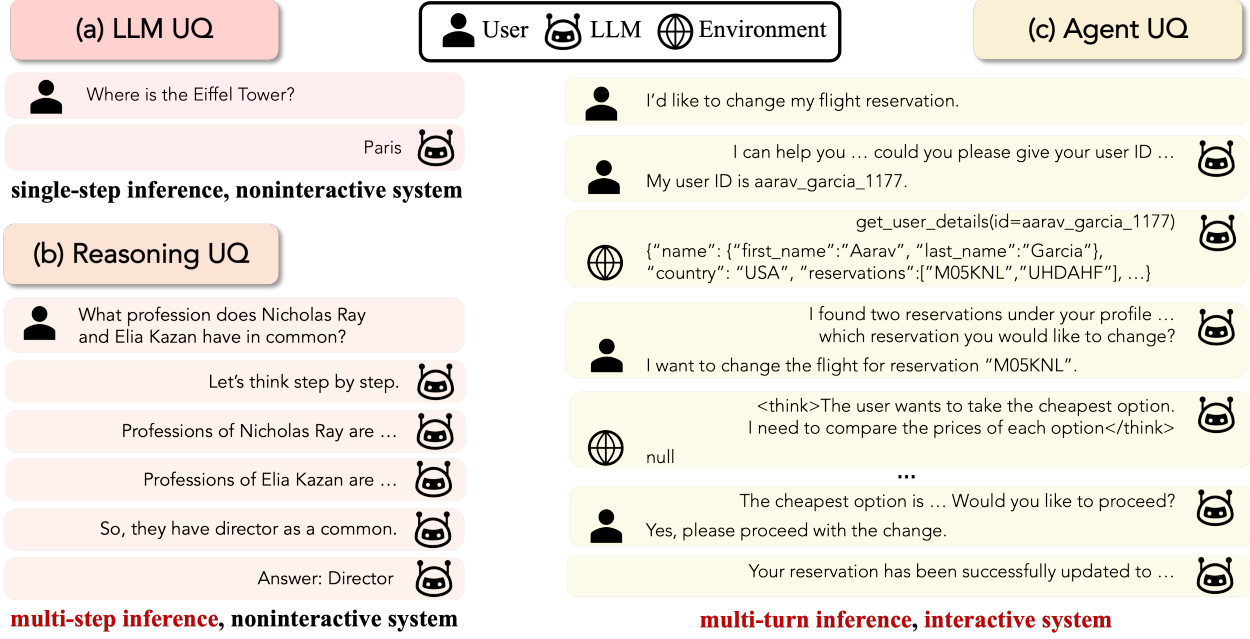


Figure 1 | **Comparison between UQ setups.** Traditional LLM UQ (a) measures the uncertainty of an answer given a question, whereas the UQ for LLM reasoning (b) expands the problem by considering multi-step responses rather than a single response. Agent UQ (c) goes further by considering continual interactions between agent and user/environment across the trajectory, making it a multi-turn, interactive inference setup (example sourced from τ^2 -bench Airline; Barres et al. (2025)).

LLM agents in this setup operate with feedback from users and system components through multi-turn interactions to complete a complex task (Figure 1, right). For example, a flight-booking agent needs to decide whether to finalize a reservation or ask follow-up questions about dates, budget, or layover preferences. Early uncertainty should prompt information-seeking actions, while subsequent dialogue and database queries can resolve ambiguities and reduce uncertainty, enabling confident execution. This deviates significantly from the setup considered in classic LLM UQ, as the agent now spans a long interaction log updated over multiple turns that involves messages from other entities, and it also gets additional information from environments. This gap raises questions about the applicability of existing UQ methods to agentic setups.

In this paper, we argue that building a trustworthy LLM-driven agentic system requires a shift in focus: from pointwise uncertainty of the final answer to structured uncertainty dynamics in an open, interactive decision process, and a new framework is necessary for this. To initiate a formal discussion, we first **provide a concrete definition and a general formulation of agent UQ** that captures broad categories of existing UQ setups (Sec. 3). Specifically, we model the agent’s problem-solving trajectory as a stochastic process over actions, observations, and environment states, represented as a simple graphical model. Then, we define both turn-level and trajectory-level uncertainty in a simple, probabilistic expression, where we show that many prior UQ problems can be cast as special cases.

Based on this foundation, we **identify four technical challenges in agent UQ** that call for actions from communities. (1) Selection of uncertainty estimator: while it is already known that existing uncertainty estimators have their own tradeoffs, agentic scaffolding amplifies these into severe limitations or hard constraints; (2) Uncertainty of heterogeneous entities: how can we derive estimates over the content generated by external entities having different underlying distributions?; (3) Modeling uncertainty dynamics in interactive systems: traditional weighted-average-style uncertainty aggregation may not be suitable to modeling uncertainty dynamics in the non-isolated, open environment; (4) Lack of fine-grained benchmarks: our pilot survey reveals that granular benchmarks, such as turn-level evaluation, are scarce for LLM agents.

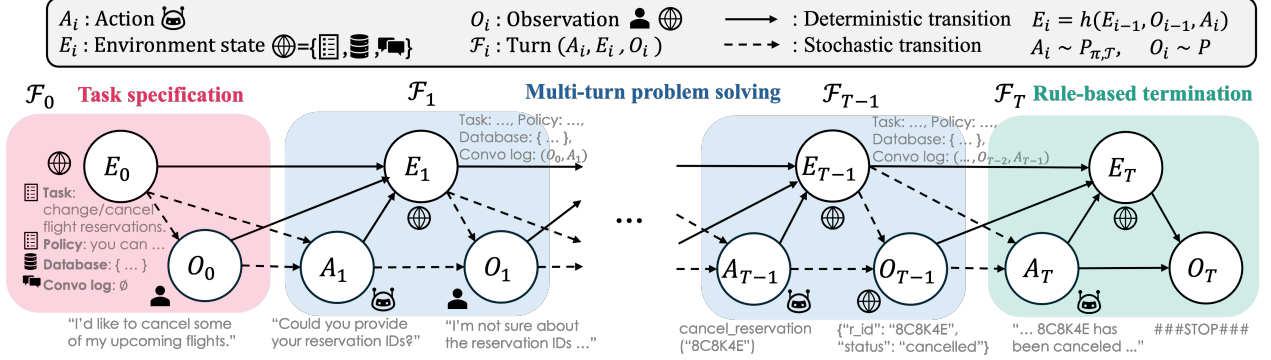


Figure 2 | **Graphical model for an agent problem-solving trajectory with examples.** Given a task specification E_0 and an initial user query O_0 , an agent spans a multi-turn trajectory characterized by a chain of action A , observation O , and environment state E . This simple abstraction describes some representative agentic prompting methods such as ReAct (Yao et al., 2022b). See Appendix B.1 for details.

Then, Sec. 5 discusses **practical implications** of agent UQ across diverse domains, including healthcare, software engineering, and robotics, as well as frontier LLM research on adaptive reasoning and post-training. Finally, we outline **open problems** specific to uncertainty modeling in agentic systems (Sec. 6). Together, we view this work as a step toward providing the field with a necessary foundation and clearer direction for future research on uncertainty-aware, reliable agentic systems.

- We cast agent UQ as a distinct problem setup, and build a foundation for that by providing a concrete definition and a general formulation that expresses many existing UQ setups as special cases.
- We crystallize the emerging challenges in agent UQ with numerical studies on a real-world benchmark τ^2 -bench (Barres et al., 2025) with GPT-4.1 (OpenAI, 2025b) and Kimi-K2.5 (Kimi Team, 2026).
- To further give actionable insights and promote discussion, we describe promising examples of agent UQ applications in various domains and also remaining open problems.

2. Related Work

UQ in LLMs. In contrast to classic UQ in machine learning (Gal and Ghahramani, 2016; MacKay, 1992; Neal, 1992), LLM UQ brings extra challenges from its computational cost and free-form outputs. Common approaches include aggregating output token probability (Aichberger et al., 2024; Duan et al., 2024; Fadeeva et al., 2024; Fomicheva et al., 2020; Malinin and Gales, 2021; Zhang et al., 2023, 2025b), measuring consistency over multiple generations (Farquhar et al., 2024; Manakul et al., 2023; Nikitin et al., 2024), verbalizing confidence (Kadavath et al., 2022; Lin et al., 2022; Yona et al., 2024), or conformal prediction (Cherian et al., 2024; Kumar et al., 2023; Quach et al., 2024). While promising, they mostly use single-step question-answering as a testbed, which limits their practical applicability to agentic setups.

UQ in reasoning and agent setups. Beyond single-step generation, researchers show growing interest in measuring the confidence of LLMs during their reasoning. This (un)certainly information has been used to improve response quality (Fu et al., 2025; Kang et al., 2025b; Wang et al., 2023) or to guide a model during inference or training (Cheng et al., 2025; Hu et al., 2024; Lightman et al., 2024; Wang et al., 2025a,b; Zhu et al., 2025). Recently, there have been attempts to model uncertainty in interactive or agentic inference (Chan et al., 2025; Duan et al., 2025; Frankel et al., 2024; Han et al., 2024; Zhang et al., 2026a,c; Zhao et al., 2025). Although aiming to model uncertainty dynamics over long-horizon generation, they *do not reflect uncertainty from different entities* (e.g., users) *nor concern the reducible nature of uncertainty in open environments* (Kirchhof et al., 2025; Suri et al., 2025), highlighting the need for a new UQ framework under agentic scaffolding.

3. A General Formulation for Agent UQ

3.1. Problem Statement and Definitions

Imagine a flight booking agent depicted in Figure 1 right. Here, the types of **action** A and **observation** O are diverse: for A , there is information-seeking, questioning for clarification, calling an API function, and thinking or updating without interactive behaviors. The observation O , on the other hand, can be a database status report, tool-calling results, or the user’s clarification. Both action and observation can be realized in flexible forms of natural language messages or structured strings, e.g., programming language.

Concretely, given a **task specification** E_0 , e.g., a flight booking, a user makes a stochastic **initial query** O_0 , and the agent first requests the user’s ID to get relevant information. Then, it calls a programmed function to get privileged information from the database. Not only does the agent interact with the user or tool, but it also reasons itself about better future actions (i.e., planning) or makes a system update without external observation. All previous interactions are stored in **environment state** E for context-grounded actions over a long horizon inference. This makes up a **trajectory** $\mathcal{F}_{\leq t}$ (chain of A , E , and O) up to t -turn which continues until the task-specific termination rule is met, e.g., when the agent submits an answer (Jimenez et al., 2024; Wei et al., 2025; Yang et al., 2018), the user generates a stop sign (Barres et al., 2025; Yao et al., 2024), or goal completion (Shridhar et al., 2021) with multiple constraints (Xie et al., 2024a).

Now, to formalize UQ in this real-world agent context, we begin with a concrete definition of the agent system in Definition 1, and introduce a graphical model for the agent trajectory in Figure 2. This can generalize many existing agent benchmark setups (Jimenez et al., 2024; Liu et al., 2024; Mialon et al., 2023; Song et al., 2023; Wei et al., 2025; Xie et al., 2024a; Yang et al., 2018; Yao et al., 2024).

Definition 1 (Stochastic Agent System). Let E_i be an environment state, a mixture of (1) a context memory of agent-user-tool interaction logs up to the i -th turn that is fully accessible to the agent, and (2) a state of the system database, which is not directly accessible to the agent—only partially observable via tool-calling. Let O_i and A_i be the i -th turn observation and action derived from distributions P and $P_{\pi, \mathcal{T}}$, respectively, where $\mathcal{T}$ indicates a tool set and π is an LLM policy. Then, under a task specification E_0 and the user’s initial query O_0 , we define the trajectory as a sequence of turns $\mathcal{F}_{\leq T} = \{(A_t, E_t, O_t)\}_{t=0}^T$ where $\mathcal{F}_0 = (E_0, O_0)$. Given a deterministic update function $h(\cdot)$, the generative process for each turn t is defined as:

$$A_i \sim P_{\pi, \mathcal{T}}(\cdot | E_{i-1}, O_{i-1}), \quad O_i \sim P(\cdot | A_i, E_i), \quad E_i = h(E_{i-1}, O_{i-1}, A_i).$$

To express this trajectory with a simple mathematical representation, we introduce a dynamic Bayesian network (Murphy, 2002) in Figure 2. With this graphical model, dependency structures of A , E , and O can be explained as follows: (1) since the environment state E_{i-1} contains the entire conversation history of $A_{\leq i-1}$ and $O_{< i-1}$ up to the $i-1$ step¹, the current action A_i depends only on E_{i-1} and O_{i-1} ; (2) the current observation O_i from the environment also depends only on A_i and E_i ; (3) the transition of the environment state ($E_{i-1} \rightarrow E_i$) can be defined as a deterministic function² $h(\cdot)$ which stacks the conversation logs up in a memory module and conducts a rule-based modification (write, update, or delete) on a previous system database state given the current action.

Under this Bayesian network, the joint probability for the trajectory can be factorized as below,

$$\begin{aligned} P(\mathcal{F}_{\leq T}) &= P(E_0, O_0) \prod_{i=1}^T P(\mathcal{F}_i | \mathcal{F}_{i-1}) \\ &= P(E_0, O_0) \prod_{i=1}^T P_{\pi, \mathcal{T}}(A_i | E_{i-1}, O_{i-1}) P(O_i | A_i, E_i), \end{aligned}$$

¹An adaptive memory structure (Xiong et al., 2025; Xu et al., 2025b) can replace this ever-growing memory.

²In general, this system transition can be also stochastic, but we consider a simplified yet still realistic scenario.